

POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

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Chapter 4: SERIES

CALCULUS 1

Faculty of Fundamental Science 1

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4.1. SERIES OF NUMBERS

Definition 4.1.1

Let $\{a_n\}$ be a sequence of real numbers (sequence of complex numbers). Then, the formal sum

$$a_1 + a_2 + \dots + a_n + \dots$$

is called an *infinite series* (or just a series) (series of complex numbers) and is denoted, for short, by the symbol $\sum_{n=1}^{\infty} a_n$ or $\sum a_n$, a_n is general term of series.

Example 4.1.1

$\sum_{n=1}^{\infty} \frac{1}{n}$ is a harmonic series; $\sum_{n=1}^{\infty} \frac{n+3}{n^2+5n+1}$.

4.1. SERIES OF NUMBERS

Definition 4.1.2

Given a series $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \cdots + a_n + \cdots$, let S_n denote its n -th partial sum:

$$S_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k.$$

If the limit exists $\lim_{n \rightarrow \infty} S_n = S$, the series is said to be *convergent* to S .

If the limit does not exist, the series is said to be *divergent*.

Example 4.1.2

- a) Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ is convergent, and find its sum.
- b) Show that the series $\sum_{n=1}^{\infty} \ln \frac{n}{n+1}$ is divergent.

4.1. SERIES OF NUMBERS

Special series

1. Geometric series $\sum_{n=1}^{\infty} p^{n-1} = 1 + p + p^2 + \cdots$, where p

is constant, it converges to $S = \frac{1}{1-p}$ if $|p| < 1$ and diverges if $|p| \geq 1$.

The sum of the first n terms is $S_n = \frac{1 - q^n}{1 - q}$.

2. The α series $\sum_{n=1}^{\infty} \frac{1}{n^\alpha} = 1 + \frac{1}{2^\alpha} + \frac{1}{3^\alpha} + \cdots$ is called a Riemann series

where α is constant, converges for $\alpha > 1$ and diverges for $\alpha \leq 1$.

The series with $\alpha = 1$ is called the harmonic series.

4.1. SERIES OF NUMBERS

Theorem 4.1.1 (Cauchy's theorem)

The series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\forall \varepsilon > 0, \exists n_0, \forall n \geq n_0, \forall p \in \mathbb{N} = \{0, 1, \dots\}$ such that

$$|S_{n+p} - S_n| < \varepsilon.$$

Theorem 4.1.2 (Necessary condition)

If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$.

4.1. SERIES OF NUMBERS

Remark:

- a) In the case $\lim_{n \rightarrow \infty} a_n \neq 0$, then we deduce that $\sum_{n=1}^{\infty} a_n$ is divergent.
- b) The series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the series $\sum_{n=n_0}^{\infty} a_n$ is also convergent for each $n_0 > 0$. It follows from

$$\sum_{n=1}^{\infty} a_n = (a_1 + a_2 + \dots + a_{n_0-1}) + \sum_{n=n_0}^{\infty} a_n.$$

Example 4.1.3

Show that the series $\sum_{n=1}^{\infty} \frac{2n^2 + 5n + 7}{n^2 + n + 3}$ is divergent.

4.1. SERIES OF NUMBERS

Theorem 4.1.3

If the series $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ are convergent, then so are the series $\sum_{n=1}^{\infty} ca_n$ (where c is a constant), $\sum_{n=1}^{\infty} (a_n + b_n)$ and $\sum_{n=1}^{\infty} (a_n - b_n)$, and

$$i) \sum_{n=1}^{\infty} ca_n = c \sum_{n=1}^{\infty} a_n;$$

$$ii) \sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n;$$

$$iii) \sum_{n=1}^{\infty} (a_n - b_n) = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n.$$

4.2. Positive Series

Definition 4.2.1

The $\sum_{n=1}^{\infty} a_n$ is called a positive series, if $a_n > 0, \forall n \in \mathbb{N}^*$ (i.e, a_n is positive terms).

Example 4.2.1

$\sum_{n=1}^{\infty} \frac{1}{n^2}; \quad \sum_{n=1}^{\infty} \frac{n+2}{2n^3+n+7}$ are positive series.

4.2. Positive Series

Theorem 4.2.1 (Comparison Test)

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms such that $a_n \leq b_n, \forall n \geq n_0$. The following holds,

- If $\sum_{n=1}^{\infty} b_n$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is also convergent.
- If $\sum_{n=1}^{\infty} a_n$ is divergent, then $\sum_{n=1}^{\infty} b_n$ is also divergent.

Example 4.2.2

Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 5n + 2021}$ converges or diverges.

4.2. POSITIVE SEQUENCE

Theorem 4.2.2 (Limit Test)

Suppose that $\sum_{n=1}^{\infty} a_n$ ⁽¹⁾ and $\sum_{n=1}^{\infty} b_n$ ⁽²⁾ are positive series and

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = M \geq 0. \text{ Then,}$$

If $0 < M < +\infty$ then either both series converge or both diverge.

If $M = 0$ and (2) converges, then (1) converges.

If $M = +\infty$ and (2) diverges, then (1) diverges.

4.2. POSITIVE SEQUENCE

Example 4.2.2

Determine convergent or divergent series

$$\sum_{n=1}^{\infty} n \sin^2 \frac{2}{n^2}.$$

4.2. POSITIVE SEQUENCE

Theorem 4.2.3 (D'Alembert's criterion)

Suppose that positive series $\sum_{n=1}^{\infty} a_n$ and let $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$. This follows that

If $L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

If $L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

If $L = 1$ the test fails.

Example 4.2.3

Determine whether series is convergent or divergent: $\sum_{n=1}^{\infty} \frac{n!}{3^n}$.

4.2. POSITIVE SEQUENCE

Theorem 4.2.4 (Cauchy's criterion)

Consider positive series $\sum_{n=1}^{\infty} a_n$ and evaluate $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = C$. Then

If $C < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

If $C > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

If $C = 1$ the test fails.

Example 4.2.4

Prove that the following series is divergent: $\sum_{n=1}^{\infty} \left(\frac{2n^2 + n}{n^2 + 5} \right)^n$.

4.2. POSITIVE SEQUENCE

Theorem 4.2.5 (Integral criterion)

Let the function $f(x)$ be continuous, positive and decreasing to 0 on $[1, \infty)$ (i.e., $\lim_{x \rightarrow +\infty} f(x) = 0$). Set $a_n = f(n)$. Then, we claim that the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x)dx$ is convergent.

Example 4.2.5

Prove that the series

$$\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}$$

is convergent if and only if $\alpha > 1$.

4.3 ALTERNATING SERIES

Definition 4.3.1

Any series whose terms alternate between positive and negative values is called an alternating series. An alternating series can be written in the form

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + \cdots$$

or

$$\sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + a_4 - \cdots,$$

where $a_n > 0$ for all positive integers n .

Example 4.3.1

$$-1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \cdots = \sum_{n=1}^{\infty} (-1)^n \frac{1}{n};$$

4.3. Alternating Series

Theorem 4.3.1 (Leibniz's criterion)

Let the alternating series of the form

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + \cdots \quad (1)$$

or

$$\sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + a_4 - \cdots \quad (2)$$

satisfy

- i) $\{a_n\}$ is a decreasing sequence. It means that $0 < a_{n+1} \leq a_n \quad \forall n \geq 1$,
- ii) $\lim_{n \rightarrow \infty} a_n = 0$.

Then, the alternating series (1) and (2) are convergent.

4.3. Alternating Series

Example 4.3.2

For each of the following alternating series, determine whether the series converges or diverges

$$i) \sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2};$$

$$ii) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n+1}.$$

Definition 4.3.2

The series $\sum_{n=1}^{\infty} a_n$ is called *absolutely convergent* if $\sum_{n=1}^{\infty} |a_n|$ converges. If

$\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} |a_n|$ diverges, then $\sum_{n=1}^{\infty} a_n$ is called *semi-convergent*.

Example 4.3.2

For each of the following series, determine whether the series converges:

a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n+3}$; b) $\sum_{n=1}^{\infty} \frac{\sin(n)}{n^2}$.

Note that if $\sum_{n=1}^{\infty} |a_n|$ is convergent then $\sum_{n=1}^{\infty} a_n$ is convergent. Inverse is not true.

4.4 INFINITE SERIES OF FUNCTIONS

Definition 4.4.1

Let $f_1(x), f_2(x), \dots, f_n(x), \dots$ be a sequence of functions defined in (a, b) . Then the infinite sum

$$\sum_{n=1}^{\infty} f_n(x) = f_1(x) + f_2(x) + \dots + f_n(x) + \dots (*)$$

is called a *infinite series of functions* defined on (a, b) .

+The function $f_n(x)$ is called *the n th term* of $(*)$.

Example 4.4.1

$$\sum_{n=1}^{\infty} \frac{x^n}{n+5};$$

$$\sum_{n=1}^{\infty} \frac{\cos nx}{n^2}.$$

4.4 INFINITE SERIES OF FUNCTIONS

Definition 4.4.1 (continuity)

- + A finite sum $S_n(x) = \sum_{k=1}^n f_k(x)(**)$ is called *n-th the sequence of partial sum* of $(*)$.
- + The point x_0 is said to be *the point of convergence* (or divergence) of the series of functions $(*)$, if the series of numbers $\sum_{n=1}^{\infty} f_n(x_0)$ converge (or diverge).
- + The set of convergence points of the series of functions is called *domain of convergence* of the infinite series of functions $(*)$.
- + The series of functions $(*)$ is called *uniformly convergent* to the function $f(x)$ on the domain D , if

$$\forall \epsilon > 0, \exists n_0, \forall n \geq n_0 : |S_n(x) - f(x)| < \epsilon \quad \forall x \in D.$$

4.4 INFINITE SERIES OF FUNCTIONS

Definition 4.4.1 (continuity)

+ If the infinite series of functions (*) converges to $f(x)$, then function

$$R_n(x) = f(x) - S_n(x)$$

is called *degree remainder n* of the series of functions (*).

+ The series of functions (*) is said to be *absolutely convergent* on the domain D , if $\sum_{k=1}^{\infty} |f_k(x)|$ converges on the domain D .

Theorem 4.4.1 (Cauchy Test)

The infinite series of functions (*) is uniformly convergent on the domain D if and only if

$$\forall \epsilon > 0, \exists n_0, \forall n \geq n_0, \forall p \in \mathbb{N} : |S_{n+p}(x) - S_n(x)| < \epsilon \quad \forall x \in D.$$

Example 4.4.2

Prove that the infinite series of functions

$$i) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{x^2 + n}; \quad ii) \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^2}{(x^2 + 1)^n}.$$

are uniformly convergent on $\mathbb{R}$.

Solution:

i) Put $S_n(x) = \sum_{k=1}^n \frac{(-1)^{k-1}}{x^2+k}$. Then

$$|S_{n+p}(x) - S_n(x)| = \left| \frac{(-1)^n}{x^2+n+1} + \frac{(-1)^{n+1}}{x^2+n+2} + \dots + \frac{(-1)^{n+p+1}}{x^2+n+p} \right| \frac{1}{x^2+n+1} < \frac{1}{n} < \epsilon \Rightarrow n > \frac{1}{\epsilon} \Rightarrow n_0 = \left[\frac{1}{\epsilon} \right] + 1.$$

ii) Similar, we can also prove that

$$|S_{n+p}(x) - S_n(x)| < \frac{x^2}{(1+x^2)^n} < \frac{1}{n} < \epsilon \Rightarrow n > \frac{1}{\epsilon} \Rightarrow n_0 = \left[\frac{1}{\epsilon} \right] + 1.$$

Theorem 4.4.2 (Weierstrass Test)

If sequence of positive constants $c_1, c_2, \dots, c_n, \dots$, can be found such that

$$i) |f_n(x)| \leq c_n; \quad n = 1, 2, 3, \dots \text{ for all } x \text{ in } D;$$

$$ii) \sum_{n=1}^{\infty} c_n \text{ converges,}$$

then $\sum_{n=1}^{\infty} f_n(x)$ is uniformly and absolutely convergent in the domain D .

Example 4.4.3

$\sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$ is absolutely and uniformly convergent in $[0, 2\pi]$ since

$$\left| \frac{\cos nx}{n^2} \right| \leq \frac{1}{n^2} \text{ and } \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges.}$$

Theorem 4.4.2 (Properties continuous)

If $f_n(x), n = 1, 2, 3, \dots$ are continuous in $[a, b]$ and if $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly to the sum $f(x)$ in $[a, b]$, then $f(x)$ is continuous in $[a, b]$.

Theorem 4.4.3 (Properties integral)

If $f_n(x), n = 1, 2, 3, \dots$ are continuous in $[a, b]$ and if $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly to the sum $f(x)$ in $[a, b]$, then

$$\int_a^b f(x)dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)dx \text{ or } \int_a^b \left\{ \sum_{n=1}^{\infty} f_n(x) \right\} dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)dx.$$

Theorem 4.4.4 (Differentiable)

If $f_n(x), n = 1, 2, 3, \dots$ are continuous and have continuous derivatives in $[a, b]$ and if $\sum_{n=1}^{\infty} f_n(x)$ converges to the sum $f(x)$ while $\sum_{n=1}^{\infty} f'_n(x)$ is uniformly convergent on $[a, b]$. Also on $[a, b]$, we have

$$f'(x) = \sum_{n=1}^{\infty} f'_n(x) \quad \text{or} \quad \frac{d}{dx} \sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} \frac{d}{dx} f_n(x).$$

4.5 Power Series

Definition 4.5.1

A power series is an infinite series of functions of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x^1 + a_2 x^2 + \cdots ,$$

$$\text{or } \sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1 (x - x_0)^1 + a_2 (x - x_0)^2 + \cdots ,$$

where x_0 is constant and the numbers $a_0, a_1, \cdots$ are called the coefficients of the power series.

Example 4.5.1

$$\sum_{n=0}^{\infty} \frac{n+1}{2n+5} x^n;$$

$$\sum_{n=1}^{\infty} \frac{1}{n} (x-2)^n.$$

4.5 Power Series

Note that (definition): A power series $\sum_{n=0}^{\infty} a_n x^n$ is called to be *convergent* at the point $x = \alpha$, if the numerical series $\sum_{n=0}^{\infty} a_n \alpha^n$ is convergent.

Theorem 4.5.1 (Abel)

If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges at the point $x_0 \neq 0$, then it converges at any point x satisfying $|x| < |x_0|$. Moreover, if it diverges at the point x_1 , then it diverges at any point x satisfying $|x| > |x_1|$.

Definition

A power series $\sum_{n=0}^{\infty} a_n x^n$ converges for all $|x| < R$ and diverges for all $|x| > R$, where the constant R is called the *radius* of convergence of the power series. For $|x| = R$, the series may or may not converge.

4.5 Power Series

Theorem 4.5.2 (Rules of finding radius of convergence)

Consider a power series $\sum_{n=0}^{\infty} a_n x^n$. If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho$ or $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \rho$ exists, then

$$R = \begin{cases} \frac{1}{\rho} & \text{if } 0 < \rho < +\infty, \\ +\infty & \text{if } \rho = 0, \\ 0 & \text{if } \rho = +\infty. \end{cases}$$

Example 4.5.2

Determine the radius of convergence and region of convergence for the following power series

$$i) \sum_{n=0}^{\infty} \frac{x^n}{n!}; \quad ii) \sum_{n=1}^{\infty} n^n x^n; \quad iii) \sum_{n=1}^{\infty} \frac{(x-2)^n}{\left(\frac{n+1}{n}\right)^{n^2}}.$$

4.5 POWER SERIES

Theorem 4.5.3 (Uniformly convergent of power series)

If $\sum_{n=0}^{\infty} a_n x^n$ has a radius of convergence $R > 0$, then it converges in $[-R', R']$ for any $R' : 0 < R' < R$.

Theorem 4.5.4 (Properties of power series)

Given a power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $R > 0$. Put

$f(x) = \sum_{n=0}^{\infty} a_n x^n$. Thus, i) the function $f(x)$ is continuous on $(-R, R)$.

$$ii) \int_0^x f(t) dt = \sum_{n=0}^{\infty} \int_0^x a_n t^n dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} \quad \forall x \in (-R, R).$$

$$iii) f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad \forall x \in (-R, R).$$

Example 4.5.3

Calculating sum of the following power series:

$$f(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n-1} \frac{x^n}{n} + \dots \quad \text{with } x \in (-1, 1).$$

Solution

The $f(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$ converges on $(-1, 1)$. Thus

$$f'(x) = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + \dots = \frac{1}{1+x} \quad \forall |x| < 1.$$

Taking the integral, we get

$$f(x) = \int \frac{dx}{1+x} = \ln(1+x) + C.$$

Replace $x = 0$ in $f(x)$, we get $f(0) = C = 0$.

Thus, $f(x) = \ln(1+x) \quad \forall x \in (-1, 1)$.

EXPANSION OF FUNCTIONS IN POWER SERIES

Which functions have power series representations? How can we find such representations? We start by supposing that is any function that can be represented by a power series

$$f(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots + a_n(x - x_0)^n + \cdots (1)$$

Let's try to determine what the coefficients a_n must be in terms of f . To begin with,, notice that if we put $x = x_0$ in (1), then all terms after the first one are 0 and we get $f(x_0) = a_0$.

we can differentiate the series in (1) term by term

$f'(x) = a_1 + 2a_2(x - x_0) + \cdots + na_n(x - x_0)^{n-1} + \cdots (2)$ and substitution of $x = x_0$ in (2) gives $f'(x_0) = a_1$.

By now you can see the pattern. If we continue to differentiate and substitute $x = x_0$, we obtain $f^{(n)}(x_0) = 2.3.4 \cdots .n.a_n = n!a_n$.

Solving this equation for the n-th coefficient a_n , we get $a_n = \frac{f^{(n)}(x_0)}{n!}$

Definition 4.5.2 (Taylor series)

If all the derivatives of f exist, then the infinite series

$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$ (3) is called a Taylor series of the function $f(x)$, although when $x_0 = 0$, it can also be referred to as a MacLaurin series.

Theorem 4.5.5

Let the function $f(x)$ have the derivatives of all orders on the neighborhood of the point x_0 and in this neighborhood if there is an $M > 0$ such that $|f^{(k)}(x)| \leq M \quad \forall k \in \mathbb{N}$, then

$$\begin{aligned} f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n &= f(x_0) + \frac{f'(x_0)}{1!} (x - x_0) + \\ &\cdots + \dots + \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n + \cdots \end{aligned}$$

SOME IMPORTANT MACLAURIN SERIES

The following series, convergent to the given function in the indicated intervals, are frequently employed in practice:

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \dots; \forall x \in \mathbb{R}$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + \dots; \forall x \in \mathbb{R}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots; \forall x \in \mathbb{R}$$

$$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^{n-1} x^n}{n} = x - \frac{x^2}{2} + \dots + (-1)^{n-1} \frac{x^n}{n} + \dots; -1 < x \leq 1$$

$$(1+x)^\alpha = 1 + \frac{\alpha x}{1!} + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} x^n + \dots; -1 < x \leq 1$$

4.6 Fourier Series

4.6.1 Trigonometric series

A series of functions of the form $\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)(*)$, is called trigonometric series, where a_n, b_n are constants for all $n = 0, 1, 2, \dots$.

Theorem 4.6.1

If $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ are absolutely convergent, then the trigonometric series $(*)$ uniformly and absolutely convergent on $\mathbb{R}$.

Theorem 4.6.2

If two sequences of positive numbers (a_n) and (b_n) are called monotonically decreasing and convergent to 0, then the trigonometric series $(*)$ converges on the domain $D = \mathbb{R} \setminus \{k2\pi : k \in \mathbb{Z}\}$.

4.6.2 Definition of Fourier Series

Let $f(x)$ be defined and integrable on $[-\pi, \pi]$. The trigonometric series has the form

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \quad (3)$$

is called the Fourier series with the Fourier coefficients a_n and b_n are defined in the forms:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx, \quad n = 0, 1, \dots \quad (4)$$

Theorem 4.6.3 (Dirichlet's theorem)

Let $f(x)$ be defined on $[-\pi, \pi]$ satisfies the following condition:

- i) $f(x)$ is periodic outside $(-\pi, \pi)$ with period 2π .
- ii) $f(x)$ and $f'(x)$ are piece-wise continuous functions.
- iii) $f(x)$ is monotone and bounded.

Then, the series (3) with Fourier coefficients converges to

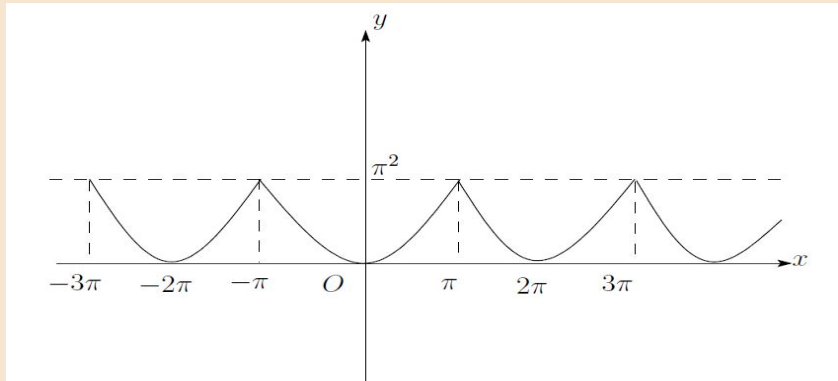
- a) $f(x)$ if x is a point of continuity.
- b) $\frac{f(x_0^+) + f(x_0^-)}{2}$ if x_0 is a point of discontinuity.

Example 4.6.1

Find the Fourier series of a periodic function with period 2π

$$f(x) = x^2 \quad \text{with} \quad -\pi \leq x \leq \pi.$$

Solution:



Hình 6.1: Graph of the function $f(x) = x^2$, $x \in [-\pi, \pi]$, period 2π

Calculating the Fourier coefficients, we have

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2\pi^2}{3}.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx = \frac{4(-1)^n}{n^2};$$

$$\text{and } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx dx = 0.$$

So the Fourier series of the function $f(x)$ has the form

$$f(x) = \frac{\pi^2}{3} - 4 \left(\cos x - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \cdots + \frac{4(-1)^{n+1} \cos nx}{n^2} + \cdots \right).$$

4.6.3 Fourier series of a function with period $2T$

Given a function $f(x)$ is defined, integrable on $[-T, T]$ and period of $2T$. By putting $t = \frac{\pi x}{T}$, we have, the function

$$g(t) = f\left(\frac{T}{\pi}t\right) \text{ periodic with period } 2\pi.$$

Thus, the Fourier series of the function $g(t)$ has the form

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + b_n \sin nt,$$

and Fourier coefficients satisfies

$$\begin{cases} a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} g(t) dt; & a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} g(t) \cos ntdt \\ b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} g(t) \sin ntdt \end{cases}, \quad n = 1, 2, \dots$$

4.6.3 Fourier series of a function with period $2T$

Note that K is called the *period* of $f(x)$, if $K > 0$ is the smallest number such that

$$f(x + K) = f(x) \quad \forall x \in D \text{ (the domain of } f(x)\text{)}.$$

The *Fourier series* of the function $f(x)$ of the form:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{\pi n x}{T} + b_n \sin \frac{\pi n x}{T} \right),$$

where the *Fourier coefficients* $a_n, b_n (n = 0, 1, 2, \dots)$ are determined by

$$\begin{cases} a_n = \frac{1}{T} \int_{-T}^T f(x) \cos \frac{n\pi x}{T} dx = \frac{1}{T} \int_0^{2T} f(x) \cos \frac{n\pi x}{T} dx \\ b_n = \frac{1}{T} \int_{-T}^T f(x) \sin \frac{n\pi x}{T} dx = \frac{1}{T} \int_0^{2T} f(x) \sin \frac{n\pi x}{T} dx. \end{cases}$$

4.6.3 Fourier series of a function with period $2l$

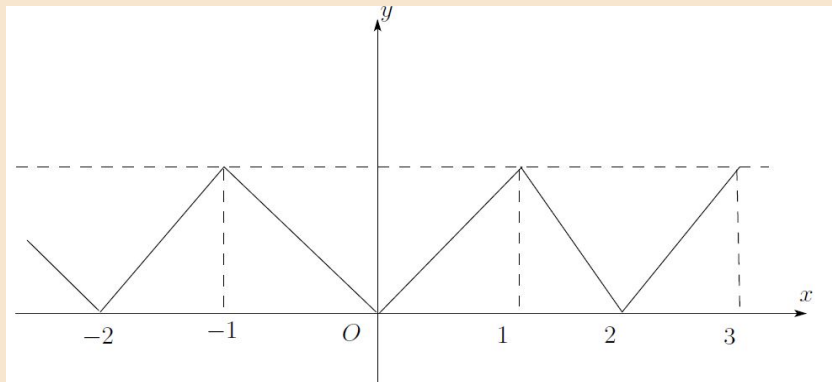
Example 4.6.2

Find the Fourier series of a periodic function with period $2l = 2$

$$f(x) = |x| \quad \text{with } -1 \leq x \leq 1,$$

and calculating the sum

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{1}{1^2} + \frac{1}{3^2} + \cdots + \frac{1}{(2n+1)^2} +$$



Hình 6.2: Graph of the function $f(x) = |x|$, $x \in [-1, 1]$, period $2l$

Calculating the Fourier coefficients with $2l = 2$, we have

$$a_0 = \frac{1}{1} \int_{-1}^1 f(x) dx = 2 \int_0^1 x dx = 1.$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos n\pi x dx = 2 \int_0^1 x \cos n\pi x dx = 2 \cdot \frac{(-1)^n - 1}{n^2 \pi^2},$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin n\pi x dx = \int_{-1}^1 |x| \sin n\pi x dx = 0.$$

Thus, the Fourier series of the function $f(x)$ has the form

$$\frac{1}{2} - \frac{2}{\pi^2} \left(\frac{\cos \pi x}{1^2} + \frac{\cos 3\pi x}{3^2} + \dots + \frac{\cos(2n+1)\pi x}{(2n+1)^2} + \dots \right).$$

Since the $f(x)$ is continuous on $\mathbb{R}$, by Dirichlet's theorem, we have

$$f(x) = \frac{1}{2} - \frac{2}{\pi^2} \left(\frac{\cos \pi x}{1^2} + \frac{\cos 3\pi x}{3^2} + \dots + \frac{\cos(2n+1)\pi x}{(2n+1)^2} + \dots \right).$$

Replacing $x = 0$ in $f(x)$, we have

$$0 = \frac{1}{2} - \frac{2}{\pi^2} \left(\frac{1}{1^2} + \frac{1}{3^2} + \dots + \frac{1}{(2n+1)^2} + \dots \right).$$

Thus,
$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{4}.$$

4.6.4 Fourier series of the function is defined on $[a, b]$, $[a, b)$, $(a, b]$, or (a, b)

Suppose for a function $f(x)$ defined on $[a, b]$, we expand the function $f(x)$ into a function $F(x)$ defined on $\mathbb{R}$ and period $2l$. Then we have the Fourier expansion of the function $F(x)$ and also the Fourier expansion of the function $f(x)$ with $x \in [a, b]$. The function progression $f(x) \rightarrow F(x)$ has the following forms:

Form 1: The the function $F(x)$ is periodic with period $2l = b - a$.

Form 2. The function $F(x)$ is even.

Form 3. The function $F(x)$ is odd.

Example 4.6.3

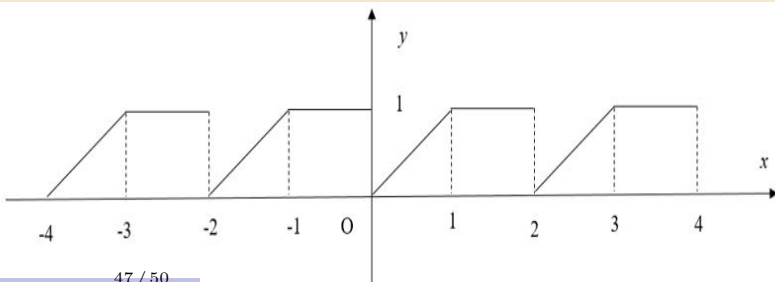
Given the function $f(x)$ defined on $(0, 2]$

$$f(x) = \begin{cases} x & \text{if } x \in (0, 1] \\ 1 & \text{if } x \in (1, 2]. \end{cases}$$

Find the Fourier series of the function $f(x)$.

Solution:

Expand the function $f(x) \rightarrow F(x)$ defined on $\mathbb{R}$ and have period $2l = 2$.



Calculating the Fourier coefficients, we have

$$a_0 = \frac{1}{l} \int_{-l}^l F(x) dx = \int_0^1 x dx + \int_1^2 1 dx = \frac{3}{2},$$

$$a_n = \frac{1}{l} \int_{-l}^l F(x) \cos \frac{n\pi x}{l} dx = \int_0^1 x \cos n\pi x dx + \int_1^2 \cos n\pi x dx = \frac{(-1)^n - 1}{n^2 \pi^2},$$

$$\begin{aligned} b_n &= \frac{1}{l} \int_{-l}^l F(x) \sin \frac{n\pi x}{l} dx \\ &= \int_0^1 x \sin n\pi x dx + \int_1^2 \sin n\pi x dx = \frac{-(-1)^n + (-1)^n - 1}{n\pi} = -\frac{1}{n\pi}. \end{aligned}$$

Since the Fourier expansion of the function $F(x)$ has the form

$$\frac{3}{4} + \sum_{n=1}^{\infty} \left(\frac{(-1)^n - 1}{n^2 \pi^2} \cos n\pi x - \frac{1}{n\pi} \sin n\pi x \right).$$

Then, the Fourier series of the function $f(x)$ on $(0, 2]$ has the form

$$\frac{3}{4} + \sum_{n=1}^{\infty} \left(\frac{(-1)^n - 1}{n^2 \pi^2} \cos n\pi x - \frac{1}{n\pi} \sin n\pi x \right).$$

Example 4.6.4

Expand

$$f(x) = \begin{cases} x & \text{if } x \in (0, 1] \\ 1 & \text{if } x \in (1, 2] \end{cases},$$

in a Fourier cosine series.

Example 4.6.5

Find the fourier sine series for a function

$$f(x) = \begin{cases} 1 - x & \text{if } x \in (0, 1] \\ 0 & \text{if } x \in (1, 2] \end{cases}.$$